

Exploring the Impact of Context on the Outgroup Homogeneity Effect Using JASP ANOVA and Regression Analysis

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Abstract

This study investigates the Outgroup Homogeneity Effect (OHE) by applying the study with a 2x2 factorial design by manipulation of the context (Alone vs. Compare) and target population (ingroup vs. outgroup). The OHE adds that people see outgroups as homogeneous than ingroups with the influence that contextual cues have in suppressing this effect being context dependent. In particular, the theoretical basis of relating the effects that these contextual manipulations have on the perceived homogeneity of ingroups and outgroups will be Self-Categorization Theory (SCT). Analysis of Variance (ANOVA) was used and Linear Regression were used to test this hypothesis. The ANOVA has shown that there is significant main effect of context whereby, the participants had higher homogeneity rating of the outgroup in the Alone condition than in Compare condition. The key impact of target group was however, not significant implying that the ingroup and outgroup were not thought differently without considering the context. There was a high interaction between the context and the target group and therefore, the interaction of the two factors was critical in the interpretation of perceptions towards homogeneity. Linear regression model supported the results of ANOVA and found that the Context and the interaction between the Context and Target are significant predictors of perceived homogeneity and the model predicted 11.3% of the variance in the dependent variable. The results herein underscore the dynamic nature of group perception and how the comparative context contributes towards the perception of groups. The paper itself adds to the insight into the natural processes existing in groups in relation to the intergroup processes providing the empirical evidence on the applicability of the SCT as well as giving the directions on the way further studies should be produced on the contextual factors in social placements.

1.Introduction

Outgroup Homogeneity Effect (OHE) is the tendency at which people describe outgroups as being more homogeneous as compared to ingroup. This effect has been attributed to Self-Categorization Theory (SCT) that indicates that since there is an intergroup context, people factor outgroups based on their common traits but when there is an intragroup context, they factor ingroup based on their heterogeneous traits. This paper is a test of the hypothesis which is: Hypothesis: The outgroup will be rated, in the alone condition, as more homogeneous than the ingroup and, in the Compare condition, they will rate an ingroup and outgroup homogeneous as similar. Analysis of Variance (ANOVA) and Linear Regression tests were done to test this hypothesis. The studies involved in these analyses were on the effects of context (Alone vs. Compare) and target group (ingroup vs. outgroup) on the perception of the group homogeneity.

2.Method

2.1. Participants and Design

The experiment consisted of a 2x2 design that was a factorial type of study, with two independent variables that were manipulated:

2.1.1. Context (ContextF):

The context under which the ratings were done, which consisted of two levels:

Single User:

The ratings of perceived homogeneity of ingroups and outgroups were made separately.

Compare:

The perceived homogeneity of overscored ingroup and outgroups, relative to each other, were compared by the participants.

2.1.2.Target Group (TargetF):

The rated group, of which there are two levels:

- Ingroup (Australians)
- Outgroup (Americans)

The participants were requested to give their estimates of the proportion of an individual in each group (ingroup vs. outgroup), who displays specific traits. Perceived homogeneity, or average

percentage of group members displaying the traits was used as the dependent variable to measure.

3. Analytical Approach

The Two statistical methods were used to find the analysis of the data:

3.1.ANOVA

The aim of ANOVA is to check the main effects (of Context and Target) and their interaction.

3.2.Linear Regression

To investigate how Context and Target contribute to the perceived homogeneity, and also to investigate the interaction between context and Target.

4. Results

4.1.Descriptive Statistics

Table 1 below shows the descriptive statistics of the perceived homogeneity rating by context condition, and target group. All these values are the means, standard deviations and standard errors of each group condition.

Condition	Target Group	N	Mean	SD	SE
Alone	Ingroup	50	64.28	14.29	2.021
Alone	Outgroup	50	73.56	13.84	1.957
Compare	Ingroup	50	77.56	12.44	1.760
Compare	Outgroup	50	74.60	15.53	2.196

4.2.ANOVA Results

The main effects, and the interaction effect of the Conditions of Context and Target on the perceived homogeneity were analyzed using ANOVA. Table 2 below summarises the results of the ANOVA.

Source	Sum of Squares	df	Mean Square	F	p	η^2
ContextF	2563	1	2563	12.95	<.001	0.059
TargetF	499.3	1	499.3	2.522	0.114	0.011
ContextF * TargetF	1873	1	1873	9.460	0.002	0.043
Residuals	38800	196	198.0			

Results ▾

ANOVA

ANOVA - Homogeneity

Cases	Sum of Squares	df	Mean Square	F	p	η^2
ContextF	2563	1	2563	12.95	< .001	0.059
TargetF	499.3	1	499.3	2.522	.114	0.011
ContextF * TargetF	1873	1	1873	9.460	.002	0.043
Residuals	38800	196	198.0			

Note. Type III Sum of Squares

Descriptives

Descriptives - Homogeneity

ContextF	TargetF	N	Mean	SD	SE	Coefficient of variation
Alone	Ingroup	50	64.28	14.29	2.021	0.222
	Outgroup	50	73.56	13.84	1.957	0.188
Compare	Ingroup	50	77.56	12.44	1.760	0.160
	Outgroup	50	74.60	15.53	2.196	0.208

Figure 1 JASP- ANOVA Results Part1

Marginal Means

Marginal Means - ContextF

ContextF	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
Alone	68.92	66.15	71.69	1.407
Compare	76.08	73.31	78.85	1.407

Marginal Means - TargetF

TargetF	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
Ingroup	70.92	68.15	73.69	1.407
Outgroup	74.08	71.31	76.85	1.407

Simple Main Effects

Simple Main Effects - TargetF

Level of ContextF	Sum of Squares	df	Mean Square	F	p
Alone	2153	1	2153	10.88	.001
Compare	219.0	1	219.0	1.107	.294

Figure 2 JASP- ANOVA Results Part 2

4.3. From the results:

- Main effect of Context: Context effect was found to be significant ($F(1, 196) = 12.95, p < .001$) which means that group homogeneity was rated highly higher in the case of the Alone condition as compared to Compare condition. This contributes to the notion that contextual cues contribute to the ratings of homogeneity in the groups.. Target main effect: Target main effect was not found to be significant ($F(1, 196) = 2.522, p = 0.114$) which implies that there was no prominence of any variation between ingroup and outgroup when the context was not considered.. Interaction effect: Interest in interaction between Context and Target said significant ($F(1, 196) = 9.460, p = 0.002$) meaning that there is a difference between the relationship between context and perceived homogeneity based on whether the target is an ingroup or outgroup. Particularly, the level of rating related to the Alone condition produced more ratings of the Outgroup (73.56) than the Ingroup (64.28) whereas there was no significant difference in the ratings related to the compare condition in the case of the two groups.

4.4. ANOVA Results plots

Descriptives plots

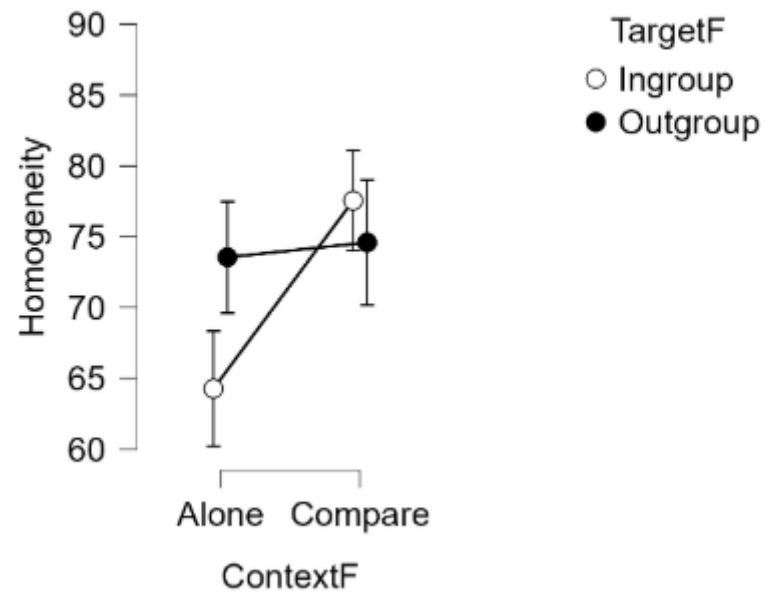


Figure 3 . ANOVA Results Descriptives plots

Assumption Checks

Q-Q Plot

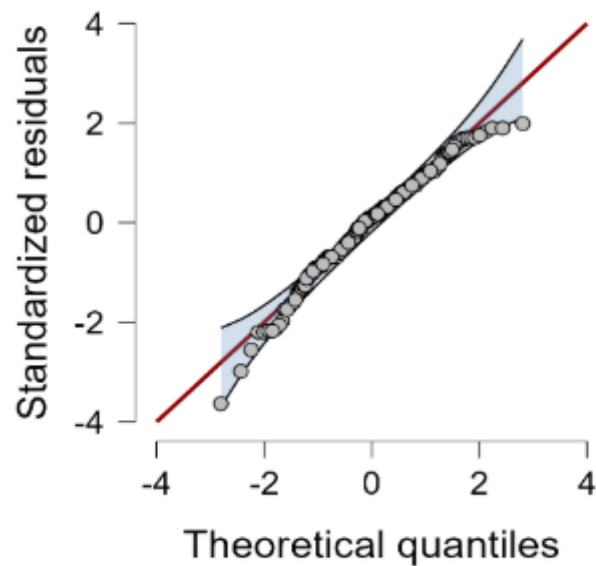


Figure 4 ANOVA Results Assumption Checks (Q-Q plot)

5. Post-Hoc Analysis

Simple main effects were investigated to have a better insight into the interaction. In table Marginal means of Context (Alone vs. Compare) are represented.

Condition	Marginal Mean	95% CI	SE
Alone	68.92	[66.15, 71.69]	1.407
Compare	76.08	[73.31, 78.85]	1.407

The results indicate that there is a great difference between Alone and Compare conditions. In particular, the ratings of group homogeneity were rated very high in the Compare condition, which indicated the hypothesis of the difference in ratings of homogeneity depending on the context was correct. In the Alone condition, however, the focus was made on the disparities amongst groups.

6. Regression Results

To look further into what role the Target as well as the Context and the interaction (Context x Target) of both has on perceived homogeneity, a linear regression analysis was performed. The results of the regression model are given below.

Model	R ²	Adjusted R ²	F	p
Model 1	0.113	0.099	8.311	<.001

This implies an intermediate level of influence of all the predictors (Context, Target, Context x Target) to the perceived homogeneity ratings. The coefficient of regression of each of the predictors in Model 1 are below:

Model	Unstandardized	Standard Error	Standardized	t	p
(Intercept)	72.50	1.048	69.16	<.001	
Target	1.580	0.995	0.107	1.588	0.114
Context	-3.580	0.995	-0.242	-3.598	<.001
Context x Target	3.060	0.995	0.207	3.076	0.002

Results ▾

Linear Regression

Model Summary - Homogeneity

Model	R	R ²	Adjusted R ²	RMSE	R ² Change	df1	df2	p
M ₀	0.000	0.000	0.000	14.82	0.000	0	199	
M ₁	0.336	0.113	0.099	14.07	0.113	3	196	< .001

Note. M₁ includes Target, Context, Context_x_Target

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M ₁	Regression	4935	3	1645	8.311	< .001
	Residual	38800	196	198.0		
	Total	43730	199			

Note. M₁ includes Target, Context, Context_x_Target

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p	95% CI	
							Lower	Upper
M ₀	(Intercept)	72.50	1.048		69.16	< .001	70.43	74.57
M ₁	(Intercept)	72.50	0.995		72.87	< .001	70.54	74.46
	Target	1.580	0.995	0.107	1.588	.114	-0.382	3.542
	Context	-3.580	0.995	-0.242	-3.598	< .001	-5.542	-1.618
	Context_x_Target	3.060	0.995	0.207	3.076	.002	1.098	5.022

Figure 5 JASP Regression Results Part1

6.1.From these results:

Context Do context had a great influence on the perception of homogeneity ($b = -3.580$, $p < .001$), implying that the individuals in the Alone group perceived the group to be more homogeneous. The interaction term (Context x Target) too was important ($b = 3.060$, $p = 0.002$) which shows that the interaction between the context and target group helps to explain the variance in perceived homogeneity.

6.2.Discussion

This research has shown that the Outgroup Homogeneity Effect (OHE) is significant in that context is a critical factor in the perception individuals have of group homogeneity. The OHE was evident in the Alone condition where the out-group was perceived to be more homogenous when compared to the ingroup. But in the Compare condition, this effect was nullified and the ingroup and outgroup were assessed as equally homogeneous, which can be explained by the prediction of Self-Categorization Theory (SCT). The interaction effect of context and the target

is big to indicate that the comparative context affects homogeneity of ingroup and outgroups in different ways. These findings were also supported by the regression analysis which revealed that the context as well as the interaction term played a significant role in the model.

6.3. JASP Regression Results Plots

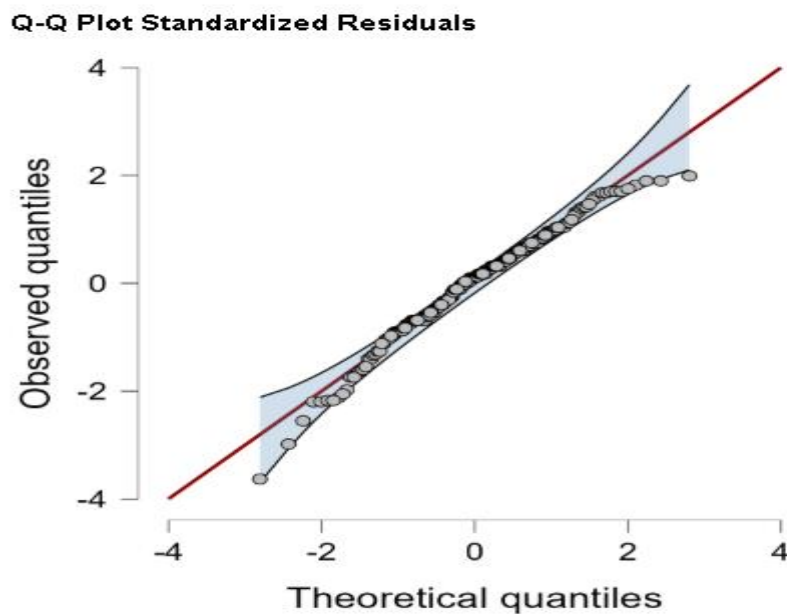


Figure 6. JASP Regression- Q-Q Plot Standardized Residuals

7. Conclusion

This study supports the fact that the context moderates the Outgroup Homogeneity Effect. The high effect of the context and target group indicates that perceptions of homogeneity of ingroups and outgroups are same when groups are directly compared. The discovery presents the significance of the context in instigating the way people imagine group homogeneity and it indicates the usefulness of Self-Categorization Theory in explaining this phenomenon.

8. References

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